Mathematical Correctness of Sufficient-Statistics OLS

sufficient-regression design note

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1 Scope and assumptions

This note gives the algebra that an IncrementalOLS, RollingOLS, weighted OLS, ridge, and exponentially forgotten OLS implementation can share. The design matrix is

$$X = \begin{bmatrix} x_1^T \\ \vdots \\ x_n^T \end{bmatrix} \in \mathbb{R}^{n\times p}, \qquad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^{n}.$$

Each row vector $x_i \in \mathbb{R}^p$ must already contain any intercept column. The formulas do not add an intercept implicitly. In the package API, this X is the internal design matrix after preprocessing: when `fit_intercept=True`, the implementation prepends a column of ones to the raw feature matrix before forming the statistics. When `fit_intercept=False`, the raw feature matrix is used as X directly.

The sufficient statistics are

$$G = X^TX = \sum_{i=1}^{n} x_i x_i^T, \qquad h = X^Ty = \sum_{i=1}^{n} x_i y_i, \qquad q = y^Ty = \sum_{i=1}^{n} y_i^2, \qquad n.$$

The matrix G is symmetric positive semidefinite. Ordinary least squares has a unique coefficient vector exactly when G is positive definite, equivalently when X has full column rank. In formulas below, inverses denote mathematical solutions. An implementation should solve linear systems using a numerically appropriate factorization and should fail loudly when the stated rank or definiteness conditions are not met.

2 Ordinary least squares from sufficient statistics

Lemma 1 (Objective expansion)**.** *For any coefficient vector $\beta \in \mathbb{R}^p$,*

$$\|y - X\beta\|_2^2 = q - 2\beta^T h + \beta^T G\beta.$$

Proof. Expand the quadratic:

$$(y - X\beta)^T(y - X\beta) = y^Ty - 2\beta^TX^Ty + \beta^TX^TX\beta = q - 2\beta^Th + \beta^TG\beta.$$

□

Theorem 1 (OLS coefficient formula)**.** *If G is positive definite, the unique ordinary least squares solution is*

$$\widehat{\beta} = G^{-1}h.$$

Proof. By the objective expansion, the gradient is

$$\nabla_\beta \left(q - 2\beta^T h + \beta^T G\beta\right) = -2h + 2G\beta.$$

Setting the gradient to zero gives the normal equation

$$G\widehat{\beta} = h.$$

Positive definiteness makes the objective strictly convex, so this stationary point is the unique minimizer. □

Corollary 1 (Residual sum of squares)**.** *For any candidate β,*

$$\mathrm{RSS}(\beta) = q - 2\beta^T h + \beta^T G\beta.$$

For the OLS solution,

$$\mathrm{RSS}(\widehat{\beta}) = q - h^T G^{-1} h.$$

Proof. The first identity is the objective expansion. For the minimizer, $G\widehat{\beta} = h$, so

$$\widehat{\beta}^T G\widehat{\beta} = \widehat{\beta}^T h = h^T G^{-1} h.$$

Substitution gives the result. □

Therefore two data sets with the same (G, h, q, n) have the same OLS objective for every β, the same OLS coefficients when the solution is unique, and the same residual sum of squares. Raw rows are not required for these quantities once the statistics are correct.

3 IncrementalOLS append formulas

Let $S_t = (G_t, h_t, q_t, n_t)$ be the statistics after t rows. When a new observation (x, y) arrives,

$$G_{t+1} = G_t + xx^T, \qquad h_{t+1} = h_t + xy, \qquad q_{t+1} = q_t + y^2, \qquad n_{t+1} = n_t + 1.$$

For a batch $X_B \in \mathbb{R}^{m\times p}$, $y_B \in \mathbb{R}^m$,

$$G_{new} = G_{old} + X_B^T X_B, \qquad h_{new} = h_{old} + X_B^T y_B, \qquad q_{new} = q_{old} + y_B^T y_B, \qquad n_{new} = n_{old} + m.$$

Theorem 2 (Correctness of append)**.** *The append formulas produce exactly the same sufficient statistics as recomputing $X^T X$, $X^T y$, and $y^T y$ from all rows after the append.*

Proof. The statistic G is a sum of row outer products. Appending (x, y) changes that sum by exactly xx^T:

$$\sum_{i=1}^{t+1} x_i x_i^T = \sum_{i=1}^{t} x_i x_i^T + x_{t+1} x_{t+1}^T.$$

The same additive argument applies to $h = \sum_i x_i y_i$, $q = \sum_i y_i^2$, and n. The batch case follows by applying the single-row argument to each row in the batch, or directly from block matrix multiplication. □

4 RollingOLS add/drop formulas

For a rolling window, the active data set is a finite multiset of rows. Suppose the current window statistics include an old observation (x_{old}, y_{old}). Dropping it gives

$$G^- = G - x_{old}x_{old}^T, \qquad h^- = h - x_{old}y_{old}, \qquad q^- = q - y_{old}^2, \qquad n^- = n - 1.$$

Adding a new observation (x_{new}, y_{new}) after the drop gives

$$G^+ = G^- + x_{new}x_{new}^T, \qquad h^+ = h^- + x_{new}y_{new}, \qquad q^+ = q^- + y_{new}^2, \qquad n^+ = n^- + 1.$$

For a fixed-size one-step slide these combine to

$$G^+ = G - x_{old}x_{old}^T + x_{new}x_{new}^T,$$
$$h^+ = h - x_{old}y_{old} + x_{new}y_{new}, \qquad q^+ = q - y_{old}^2 + y_{new}^2, \qquad n^+ = n.$$

Theorem 3 (Correctness of add/drop)**.** *If the dropped observation is in the active window, the add/drop formulas produce exactly the sufficient statistics of the new active window.*

Proof. The active-window statistic G is the sum of xx^T over exactly the rows in the window. Removing one row subtracts that row's summand, and adding one row adds the new summand. The same summand-by-summand argument applies to h, q, and n. □

Rolling deletion cannot be inferred from (G, h, q, n) alone: the implementation must know the departing row, or at least its stored contributions xx^T, xy, and y^2. This is a mathematical requirement, not an implementation preference.

5 Weighted OLS

Let $W = \mathrm{diag}(w_1, \ldots, w_n)$, with $w_i \geq 0$. The weighted least squares objective is

$$L_w(\beta) = (y - X\beta)^T W (y - X\beta) = \sum_{i=1}^{n} w_i (y_i - x_i^T\beta)^2.$$

Define weighted sufficient statistics

$$G_w = X^T W X = \sum_{i=1}^{n} w_i x_i x_i^T, \qquad h_w = X^T W y = \sum_{i=1}^{n} w_i x_i y_i, \qquad q_w = y^T W y = \sum_{i=1}^{n} w_i y_i^2.$$

If G_w is positive definite, then

$$\widehat{\beta}_w = G_w^{-1} h_w.$$

Proof. The objective expands as

$$L_w(\beta) = q_w - 2\beta^T h_w + \beta^T G_w \beta.$$

The gradient is $-2h_w + 2G_w\beta$. Positive definiteness gives a strictly convex objective, so $G_w\widehat{\beta}_w = h_w$ has the unique minimizer. □

Equivalently, weighted OLS is unweighted OLS on transformed rows $\sqrt{w_i}x_i$ and transformed responses $\sqrt{w_i}y_i$. An incremental weighted append with observation weight w is

$$G_{new} = G + wxx^T, \qquad h_{new} = h + wxy, \qquad q_{new} = q + wy^2.$$

Weighted rolling add/drop uses the same signed contributions. Negative weights do not define a least-squares norm and should be rejected.

6 Ridge regression

Let $R \in \mathbb{R}^{p\times p}$ be a symmetric positive semidefinite penalty matrix. Ridge solves

$$\widehat{\beta}_R = \arg\min_{\beta} \left[\|y - X\beta\|_2^2 + \beta^T R\beta \right].$$

Using the same sufficient statistics,

$$\|y - X\beta\|_2^2 + \beta^T R\beta = q - 2\beta^T h + \beta^T (G+R)\beta.$$

If $G + R$ is positive definite, then

$$\widehat{\beta}_R = (G+R)^{-1}h.$$

Proof. The gradient of the penalized objective is

$$-2h + 2(G+R)\beta.$$

The normal equation is $(G+R)\widehat{\beta}_R = h$. Positive definiteness of $G+R$ makes the penalized objective strictly convex, so the solution is unique. □

Scalar ridge with all coefficients penalized uses $R = \lambda I_p$, $\lambda \geq 0$. If the intercept should not be penalized, the design matrix should place the intercept in a known column and use a penalty matrix such as

$$R = \operatorname{diag}(0, \lambda, \ldots, \lambda).$$

This convention must be explicit because it changes the estimator.

7 Exponential forgetting

Let $\alpha \in (0, 1]$ be a forgetting factor. At time t, define exponentially weighted statistics

$$G_t = \sum_{i=1}^{t} \alpha^{t-i} x_i x_i^T, \qquad h_t = \sum_{i=1}^{t} \alpha^{t-i} x_i y_i, \qquad q_t = \sum_{i=1}^{t} \alpha^{t-i} y_i^2.$$

The recurrence for a new observation (x_t, y_t) is

$$G_t = \alpha G_{t-1} + x_t x_t^T, \qquad h_t = \alpha h_{t-1} + x_t y_t, \qquad q_t = \alpha q_{t-1} + y_t^2.$$

The effective weight mass follows

$$s_t = \alpha s_{t-1} + 1, \qquad s_0 = 0,$$

but s_t is not an unweighted row count when $\alpha < 1$.

Theorem 4 (Correctness of exponential forgetting)**.** *The recurrence above maintains the exponentially weighted statistics defined by the closed-form sums. If G_t is positive definite, the forgotten OLS coefficient vector is*

$$\widehat{\beta}_t = G_t^{-1} h_t.$$

Proof. Assume the closed form holds at time $t-1$. Then

$$\alpha G_{t-1} + x_t x_t^T = \alpha \sum_{i=1}^{t-1} \alpha^{t-1-i} x_i x_i^T + x_t x_t^T = \sum_{i=1}^{t} \alpha^{t-i} x_i x_i^T.$$

The same induction proves the formulas for h_t and q_t. The coefficient formula then follows from the weighted OLS theorem with weights α^{t-i}. □

With both external weights and forgetting, a new weighted observation (x_t, y_t, w_t) contributes

$$G_t = \alpha G_{t-1} + w_t x_t x_t^T, \qquad h_t = \alpha h_{t-1} + w_t x_t y_t, \qquad q_t = \alpha q_{t-1} + w_t y_t^2.$$

8 Summary

Model or operation	Statistic update or solve
OLS	$\widehat{\beta} = G^{-1}h$ when $G \succ 0$.
Append	Add xx^T, xy, y^2, and one row count.
Rolling drop	Subtract the departing row's xx^T, xy, and y^2.
Weighted OLS	Use $G_w = \sum_i w_i x_i x_i^T$, $h_w = \sum_i w_i x_i y_i$, $q_w = \sum_i w_i y_i^2$.
Ridge	Solve $(G+R)\widehat{\beta} = h$.
Forgetting	Scale previous statistics by α, then append the new contribution.